Varun Kushwah

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SUMMARY

Computer Science undergraduate focused on Java backend engineering. I build secure, concurrent services with Spring Boot, WebSockets and Redis, and I enjoy data structures, distributed systems fundamentals and clean, testable design. I coordinate Java and DSA training for a 20+ member campus tech club and have solved 500+ problems on LeetCode.

EDUCATION

KIET Group of Institutions **2024 – 2028**
B.Tech in Computer Science & Engineering | CGPA 8.27 / 10.0 Ghaziabad, India
New St. Stephens Public School CBSE XII: 80% | CBSE X: 82% 2021 – 2023 | Agra, India

TECHNICAL SKILLS

Languages: Java (17/21/25), SQL, TypeScript, JavaScript, Python
Backend: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST APIs, WebSockets
Databases: PostgreSQL, MySQL, MongoDB, Redis
Tools: Git, Docker, Postman, Linux/Bash, Maven, Gradle, CI/CD, Prometheus, Grafana
Core: Data Structures & Algorithms, OOP & Design, Concurrency, Microservices and Distributed Systems Basics

EXPERIENCE & LEADERSHIP

Java & DSA Coordinator, Technical Mentor **Aug 2024 – Present**
devup, College Tech Club Campus

- Lead the Java curriculum and hands-on backend modules for 20+ active members, preparing them for internships and technical assessments.
- Run weekly problem-solving sessions on trees, graphs and dynamic programming, improving juniors' average coding speed and accuracy by 30%.
- Conduct code reviews and Git workflows, promoting clean architecture, OOP principles and test-driven practices across cohort projects.

PROJECTS

MentorSync | LeetCode & Codeforces LMS **Live Demo | GitHub**
Java 21, Spring Boot 4, React 19, TypeScript, Redis, MongoDB, WebSockets, Resilience4j, Docker

- Built a two-sided LMS for bootcamps that automates LeetCode and Codeforces tracking, assignment distribution and real-time leaderboards.
- Wrote a verification engine on LeetCode GraphQL/REST APIs and regex parsing to validate submissions, removing manual grading.
- Added a nightly Redis cache-warming worker with atomic distributed locks (setIfAbsent), STOMP WebSockets and Resilience4j circuit breakers.

W2W Share | Offline Encrypted P2P File Transfer **Live Demo | GitHub**
Java 25, Spring Boot 4.1, WebRTC, Bouncy Castle, Virtual Threads, ZXing, Docker, Prometheus

- Engineered a peer-to-peer transfer platform on WebRTC data channels over UDP that bypasses central servers and needs no internet.
- Built a Virtual Threads and WebSocket signaling broker supporting 10k pooled connections, with ZXing QR pairing and UDP subnet discovery.
- Designed a 64KB chunk streaming protocol with AES-256-GCM encryption and backpressure, moving up to 50GB with under 50MB of RAM.

Stateless Authentication & RBAC Service **GitHub**
Java, Spring Boot, Spring Security, JWT, OAuth 2.0, PostgreSQL, Docker

- Designed a modular stateless auth service with JWT access/refresh token rotation and OAuth 2.0 social sign-in, removing server-side session state.
- Configured custom Spring Security filter chains with role-based access control across 15+ REST endpoints, with integration tests in Postman.

ACHIEVEMENTS

- **500+ problems solved on LeetCode** across arrays, strings, trees, dynamic programming and graphs.
- **Top 105 nationally, Google Solution Challenge 2025**, out of thousands of submissions.